MOHAMMED NAFEES H

Robotics & AI Engineering Student
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PROFESSIONAL SUMMARY

B.Tech Robotics & Artificial Intelligence student at SASTRA University, Class of 2028, with hands-on experience in robotics, embedded systems, computer vision, and AI backend development. Built a 4x4 Arduino-Python autonomous navigation robot, a 3-class FastAPI/Next.js fruit quality grading system, and a 3-stage YOLO/OpenCV/BLIP image-analysis backend for a larger Visage project. Skilled in ROS 2, robotics controls, Raspberry Pi, Pi Camera, OpenCV, Python, C/C++, Arduino, Linux, sensors, encoders, hardware bring-up, and model integration.

EDUCATION

SASTRA Deemed University | B.Tech, Robotics & Artificial Intelligence | Class of 2028 | Tamil Nadu, India

TECHNICAL SKILLS

- Programming: Python, C/C++, Arduino, Linux shell scripting, JSON, Git
- Robotics & Embedded: ROS 2, robotics controls, mobile robots, IR sensors, encoders, motor drivers, Raspberry Pi 5, Raspberry Pi Zero 2 W, Pi Camera, sensor/actuator interfacing, hardware debugging
- AI & Computer Vision: OpenCV, YOLOv8, Ultralytics, TensorFlow/Keras, CLIP, BLIP, Hugging Face Transformers, image classification, object detection, scene captioning
- Software & Web: FastAPI, Next.js, Tkinter, Streamlit, REST/JSON APIs, VS Code, Ubuntu 24.04
- Tools: CoppeliaSim, Fusion 360, EasyEDA/KiCad basics

EXPERIENCE

Confidential Industrial Robotics Project | Robotics Engineering Contributor | ROS 2, Controls, Embedded Integration
Project details withheld due to signed confidentiality / NDA obligations.
- Support ROS 2 based robotics workflows, controls-oriented development, embedded integration, sensor/actuator evaluation, and hardware coordination under professional confidentiality constraints.
- Coordinate component identification, evaluation, and procurement support for a project hardware budget of approximately Rs. 10 lakh.
- Work across software, electronics, and mechanical constraints for deployment-oriented industrial robotics tasks.

PROJECTS

Hadi Al Hajatron | Arduino, Python, Tkinter, IR Sensors, Encoders, A* Path Planning
GitHub: https://github.com/mohammednafees1007-hub/Hadi-Al-Hajatron
- Built a 4x4 Arduino-Python autonomous grid-navigation robot with 3 IR obstacle sensors, 60 cm calibrated movement steps, live Tkinter mapping, JSON map export, and A* route planning.

FruitVision AI - Fruit Quality Detector for SMS | FastAPI, YOLOv8x, Keras, CLIP, Next.js
GitHub: https://github.com/mohammednafees1007-hub/fruit-quality-detector-for-sms
- Built a quality-first fruit grading web system with upload/camera workflows, YOLOv8x fruit-region detection, a 3-class Keras quality model, A/B/C grade mapping, and annotated result output.

AI Image Analyzer | Python, YOLOv8n, OpenCV, BLIP, Transformers
GitHub: https://github.com/mohammednafees1007-hub/ai-image-analyzer
- Built a 3-stage backend image-analysis pipeline for a larger Visage project with YOLOv8n object detection, 13 allowed object classes, privacy-safe face tokens, BLIP captions, and 3-field JSON output.

Quadruped Robot | Inverse Kinematics, Robotics Controls, Embedded Integration
- Built inverse-kinematics motion logic for leg movement, connecting control theory with practical actuation for stable quadruped locomotion.

Schooling-Era Robotics Projects | Arduino, Sensors, Motor Drivers
- Completed early Arduino robotics prototypes involving sensors, motor drivers, wiring, control experiments, and debugging.

LEADERSHIP & INVOLVEMENT

Robotics Club Member | SASTRA Deemed University
- Active in robotics learning, project building, experimentation, and applied technical development.

ENGINEERING HIGHLIGHTS

- Practical builder across robotics software, electronics, mechanical integration, AI models, and user-facing tools.
- Comfortable with Linux robotics environments, hardware bring-up, sensor debugging, serial communication, rapid prototyping, and deployment-oriented project work.
- Strong project evidence through public GitHub repositories, runnable local applications, backend pipelines, and real hardware workflows.